STACKS, SHEAVES ON ALGEBRAIC STACKS: PROPOSITION-QUASI-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2350-2357 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2359-2452, the complete author proof (92 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_algebraic-stacks-sheaves.json (pb=3913, stmt 2350-2357, proof 2359-2452).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition lemma-2-morphisms-presheaves: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 440–454. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-characterize-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1798–1805. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-compare-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2263–2274. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-stackification-quasi-coherent: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2197–2207. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 044Q: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2452–2452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 044S: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2452–2452. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 044T: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2452–2452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let (U, R, s, t, c) be a groupoid in algebraic spaces over S. Let $\mathcal{X} = [U/R]$ be the quotient stack. The category of quasi-coherent modules on $\mathcal{X}$ is equivalent to the category of quasi-coherent modules on (U, R, s, t, c).*

Proof. We will construct quasi-inverse functors

$$QCoh(\mathcal{O}_\mathcal{X}) \longleftrightarrow QCoh(U, R, s, t, c).$$

where $QCoh(U, R, s, t, c)$ denotes the category of quasi-coherent modules on the groupoid (U, R, s, t, c).

Let $\mathcal{F}$ be an object of $QCoh(\mathcal{O}_\mathcal{X})$. Denote $\mathcal{U}$, $\mathcal{R}$ the categories fibred in groupoids corresponding to U and R. Denote x the (defining) object of $\mathcal{X}$ over U. Recall that we have a 2-commutative diagram

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{s} & \mathcal{U} \\ {\scriptstyle t}\downarrow & & \downarrow{\scriptstyle x} \\ \mathcal{U} & \xrightarrow{x} & \mathcal{X} \end{array}$$

See Groupoids in Spaces, Lemma 044S. By Lemma lemma-2-morphisms-presheaves the 2-arrow inherent in the diagram induces an isomorphism $\alpha : t^*x^*\mathcal{F} \to s^*x^*\mathcal{F}$ which satisfies the cocycle condition over $\mathcal{R} \times_{s,\mathcal{U},t} \mathcal{R}$; this is a consequence of Groupoids in Spaces, Lemma 044T. Thus if we set $\mathcal{G} = x^*\mathcal{F}|_{U_{étale}}$ then the equivalence of categories in Lemma lemma-compare-quasi-coherent (used several times compatibly with pullbacks) gives an isomorphism $\alpha : t^*_{small}\mathcal{G} \to s^*_{small}\mathcal{G}$ satisfying the cocycle condition on $R \times_{s,U,t} R$, i.e., $(\mathcal{G}, \alpha)$ is an object of $QCoh(U, R, s, t, c)$. The rule $\mathcal{F} \mapsto (\mathcal{G}, \alpha)$ is our functor from left to right.

Construction of the functor in the other direction. Let $(\mathcal{G}, \alpha)$ be an object of $QCoh(U, R, s, t, c)$. According to Lemma lemma-stackification-quasi-coherent the stackification map $[U/_pR] \to [U/R]$ (see Groupoids in Spaces, Definition 044Q) induces an equivalence of categories of quasi-coherent sheaves. Thus it suffices to construct a quasi-coherent module $\mathcal{F}$ on $[U/_pR]$.

Recall that an object $x = (T, u)$ of $[U/_pR]$ is given by a scheme T and a morphism $u : T \to U$. A morphism $(T, u) \to (T', u')$ is given by a pair (f, r) where $f : T \to T'$ and $r : T \to R$ with $s \circ r = u$ and $t \circ r = u' \circ f$. Let us call a *special morphism* any morphism of the form $(f, e \circ u' \circ f) : (T, u' \circ f) \to (T', u')$. The category of (T, u) with special morphisms is just the category of schemes over U.

With this notation in place, given an object (T, u) of $[U/_pR]$, we set

$$\mathcal{F}(T, u) := \Gamma(T, u^*_{small}\mathcal{G}).$$

Given a morphism $(f, r) : (T, u) \to (T', u')$ we get a map

$$\begin{aligned}
\mathcal{F}(T', u') &= \Gamma(T', (u')^*_{small}\mathcal{G}) \\
&\to \Gamma(T, f^*_{small}(u')^*_{small}\mathcal{G}) = \Gamma(T, (u' \circ f)^*_{small}\mathcal{G}) \\
&= \Gamma(T, (t \circ r)^*_{small}\mathcal{G}) = \Gamma(T, r^*_{small}t^*_{small}\mathcal{G}) \\
&\to \Gamma(T, r^*_{small}s^*_{small}\mathcal{G}) = \Gamma(T, (s \circ r)^*_{small}\mathcal{G}) \\
&= \Gamma(T, u^*_{small}\mathcal{G}) \\
&= \mathcal{F}(T, u)
\end{aligned}$$

where the first arrow is pullback along f and the second arrow is α. Note that if (f, r) is a special morphism, then this map is just pullback along f as $e^*_{small}\alpha = \mathrm{id}$ by the axioms of a sheaf of quasi-coherent modules on a groupoid. The cocycle condition implies that $\mathcal{F}$ is a presheaf of modules (details omitted). We see that the restriction of $\mathcal{F}$ to $(Sch/T)_{fppf}$ is quasi-coherent by the simple description of the restriction maps of $\mathcal{F}$ in case of a special morphism. Hence $\mathcal{F}$ is a sheaf on $[U/_pR]$ and quasi-coherent (Lemma lemma-characterize-quasi-coherent).

We omit the verification that the functors constructed above are quasi-inverse to each other. $\square$